

University of Houston
Cullen College of Engineering
Engineering Technology Department

Mechanical Engineering Technology

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MECT 3155

STRENGTH MATERIALS LAB

Strain Gauge Trainer 1 & 2

LAB Assignment # 06

Names: Natalie Zeledon

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Kader Adiawiakoye

PSIDs: 2136551

8016500

Abstract

This experiment investigated the operation of resistance strain gauges under bending, torsion, and tensile loading using a Strain Gauge Trainer. Quarter-, half-, and full-bridge configurations were examined to compare their sensitivity and strain measurements. Recorded readings were compared with theoretical calculations to explore the relationships between load, stress, and strain. The bending test was used to investigate Young's modulus, the torsion test examined shear and direct strain, and the tensile tests explored longitudinal strain, transverse strain, and Poisson's ratio. The experiment aimed to develop an understanding of strain-gauge applications and the factors affecting measurement accuracy.

Literature Review

Resistance strain gauges measure small deformations in materials subjected to mechanical loading. A metallic foil gauge bonded to a specimen stretches or compresses with its surface, producing a small change in electrical resistance. The gauge factor relates the fractional resistance change to the strain experienced by the gauge. Because these changes are small, a Wheatstone bridge circuit converts them into a measurable output voltage.

Quarter-, half-, and full-bridge configurations use one, two, and four active strain gauges. With appropriate gauge placement and wiring, a half bridge can produce approximately twice the output of a quarter bridge, while a full bridge can produce approximately four times the output. The configuration and gauge orientation determine sensitivity and whether bending, axial, or torsional strain is measured. Temperature changes can also affect gauge resistance. Suitable half- and full-bridge arrangements, or an unstrained temperature-compensating gauge exposed to the same temperature as the active gauge, help reduce this measurement error.

A real-world application of strain measurement is structural health monitoring of bridges. Engineers install strain gauges on structural members to measure their response to loading and compare the measurements with analytical predictions. Permanent installations support long-term monitoring, while temporary installations can help assess conditions during repairs (**Micro-Measurements, n.d.**) This application connects the laboratory's comparison of measured and calculated strains to the evaluation of real structures.

Experimental Procedure

1. Bridge Configurations and Voltage Scaling (Bending System)

The experiment was performed using a Strain Gauge Trainer to investigate strain under bending, torsion, and tensile loading. Testing began with the bending system shown below, where loads were applied to the beam and the corresponding strain and bridge output voltage were recorded for different bridge configurations.

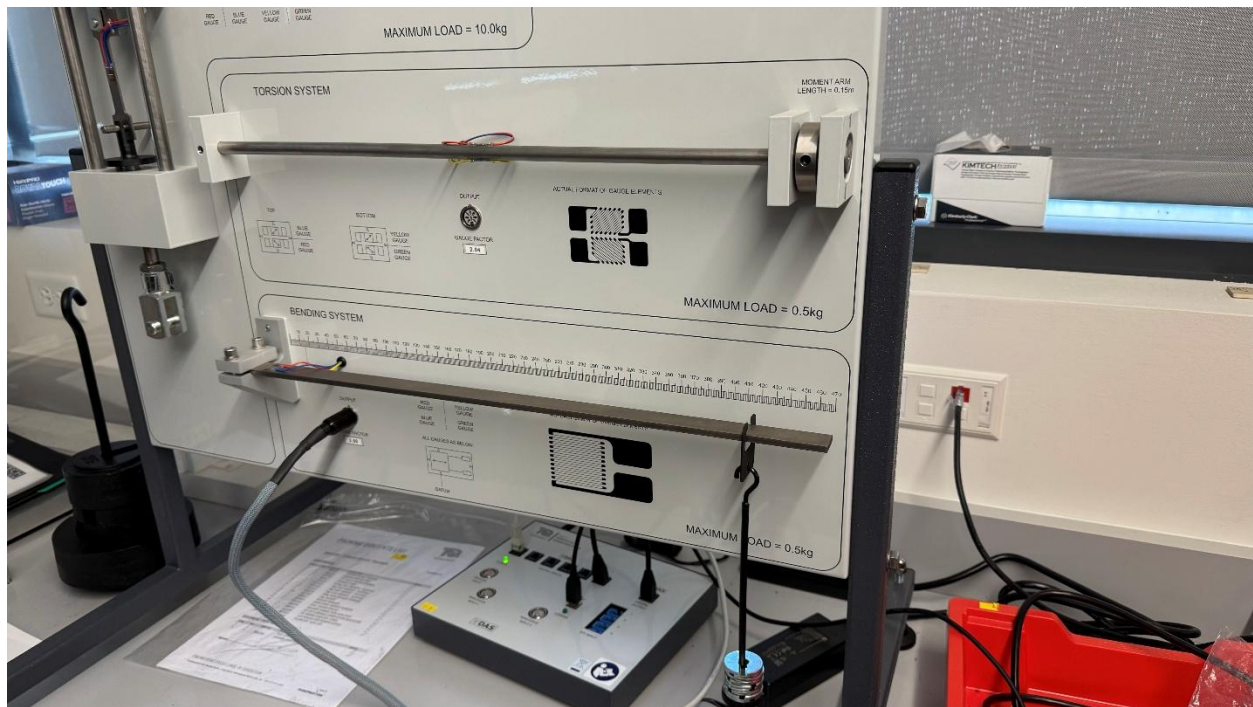


Figure 2 1 Strain Gauge Trainer Setup

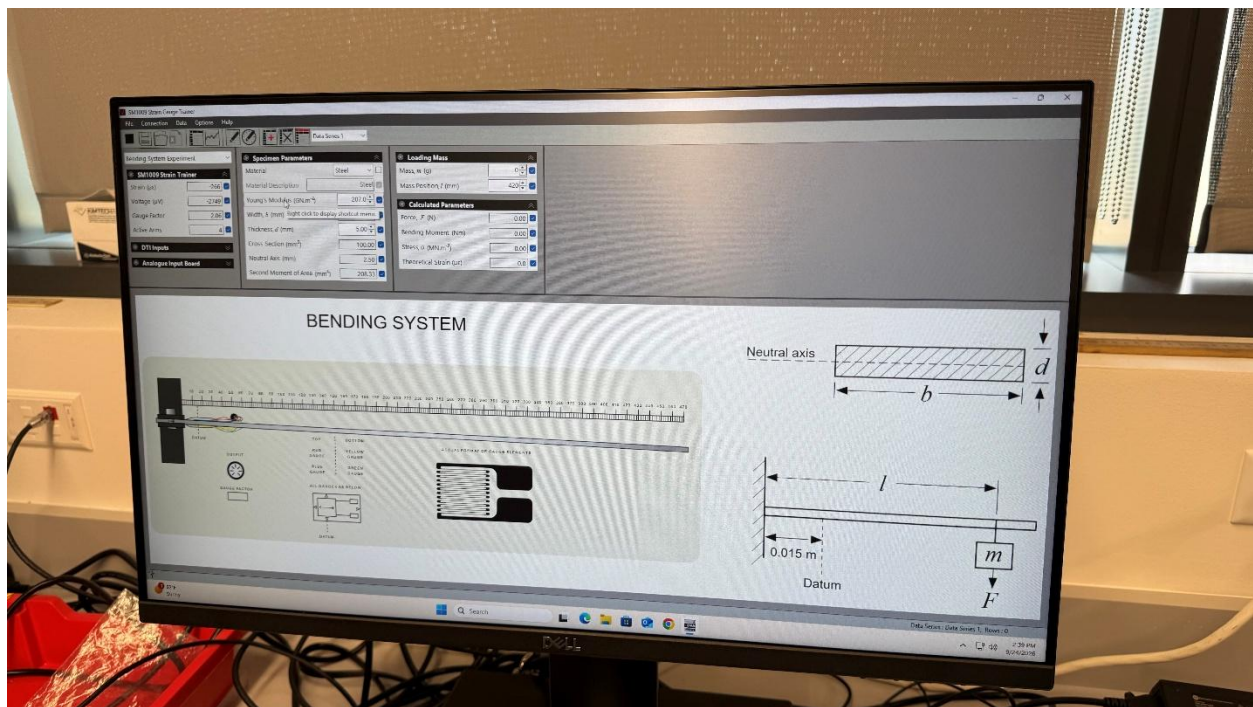


Figure 2 2 Bending system setup

The torsion measurements closely agreed with theoretical predictions, whereas the bending and tensile tests showed greater differences. The measured Young's modulus was approximately 170 GPa, below the reference value of 207 GPa. The tensile tests demonstrated longitudinal extension and transverse contraction, although the calculated Poisson's ratio requires verification of the bridge scaling before

being accepted as a material property. Overall, the experiment connected electrical strain measurements with mechanics-of-materials calculations and highlighted the importance of correct bridge configuration and data interpretation.

A quarter bridge configuration was established by wiring the red gauge to the strain display. Dummy plugs of the correct specification were placed into three sockets. The instrument was powered on. The configuration switch was set to Act = 1. The gauge factor control was set equal to the calibration value annotated on the hardware chassis. A 420 mm on the cantilever beam, a knife-edge hanger was applied. We waited for one minute for stabilization and re-set the strain display using the tare control button. A 10 g weight hanger was hung onto the beam. Further weights were added to reach a load of 50 g. The bridge output voltage and the microstrain ($\mu\epsilon$) reading was noted down. These recordings could be taken manually or automatically (using VDAS). The weights were added in intervals of 50 g up to a total weight of 500 g. For each weight, both the voltage and strain were measured.

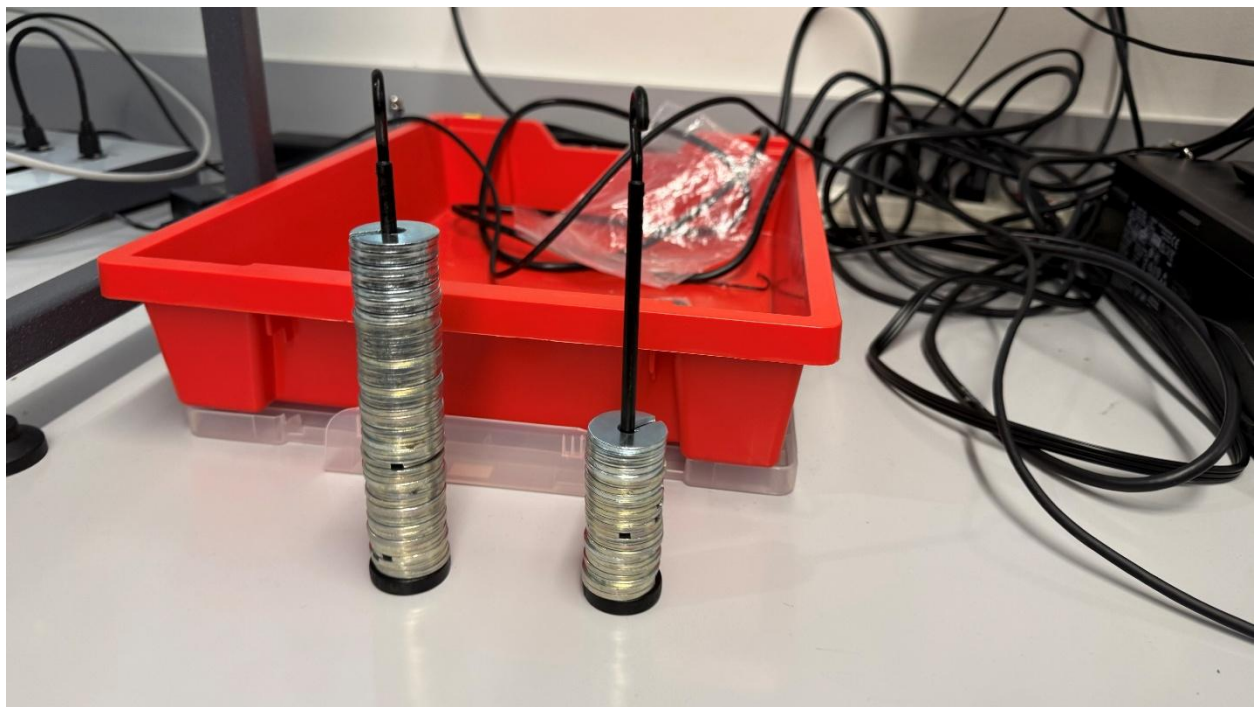


Figure 2 3 Wights added example

The procedure was then repeated for a Half Bridge 1 (Opposite Arms) configuration. The red and blue gauges were. The display was set to Act = 2. Repeating the process once more for a Half Bridge 2 (Adjacent Arms) configuration, this time with the green gauges where still set to Act = 2. We repeated the process for a Full Bridge configuration. The red and blue gauge were attached to one pair of arms, and the yellow and green gauge were attached to the other pair was set to Act = 4.

2. Analysis of Cantilever Bending Strength

The width (b) and thickness (d) of the steel cantilever beam specimen were measured using a vernier caliper. All bending system strain gauges wired to a Full Bridge configuration and knife-edge hanger position was placed at 420mm of the beam. The system was given a minute to settle and then tared to

zero. The loads were introduced from 0 g to 350 g at set increments of 50 g. The values of microstrain were recorded as the loads were applied.

3. Torsional Shear Analysis (Torsion System)

The blue strain gauge from the solid circular shaft was wired as a quarter bridge $A_{ct} = 1$; a moment arm was fastened to the end of the torsion bar, and the circuit was torn after a warm-up window of 1 minute. A load of 500 g was applied on the torque arm, and the magnitude and sign (+/-) of the microstrain was read off the gage. The weights were removed, and the procedure was duplicated for the red, yellow, and green gauges to specify the tension/compression directions. Combined loading was simulated by wiring the red/green gauges ('tensile twist') and blue/yellow gauges ('compressive twist') into opposite bridge arms and completing a Full Bridge. Microstrain was observed for the various load states of 0 g, 250 g, and 500 g.

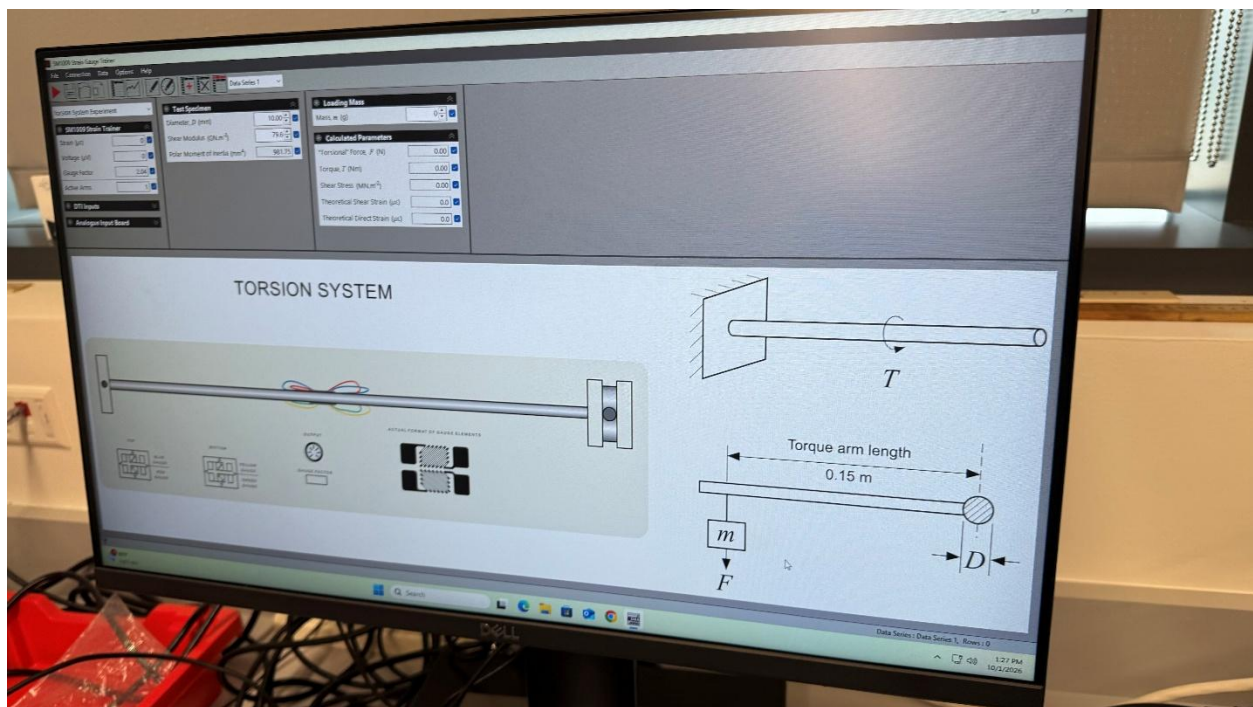


Figure 2 4 Torsion system setup

4. Axial Tension and Transverse Strain (Tension System)

A vernier caliper was employed to take a measurement of the width of the steel tensile specimen across the cross-section and of its thickness. Wire up the 2 longitudinal red and yellow strain gauges as a half bridge $A_{ct} = 2$ to detect pure axial elongation. A weight hanger of 500 g attached to the specimen base was used. Mass was added in 1 kg increments up to 10 kg and the tensile microstrain observed on the digital display were recorded. The weights were removed, and the procedure was repeated, but the transverse blue and green gauges were used as a half bridge $A_{ct} = 2$ to compensate for the effects of compressive lateral contraction. Lastly, all four tension system gauges were wired in parallel at once as a Full Bridge $A_{ct} = N$ setup, and the strain measurements were taken over the same 0-10kg load curve.

Results and Data Analysis

Experiment 1 examined how the strain-gauge bridge configuration affected the measured strain and output voltage. Loads were applied incrementally at a fixed position on the bending beam, and readings were recorded for quarter-, half-, and full-bridge connections. The results were compared to assess how changing the number of active bridge arms affected the electrical output for similar measured strains. Strain calculated from the output voltage was also compared with the instrument's displayed strain to check their agreement.

Table 1 Quarter bridge

Load (g)	Strain Reading ($\mu\epsilon$)	Output (μV)	Calculated Strain ($\mu\epsilon$)
0.	0.000	0.000	0.000
50.	12.000	32.000	12.427
100.	27.000	70.000	27.184
150.	41.000	107.000	41.553
200.	55.000	142.000	55.146
250.	69.000	180.000	69.903
300.	83.000	215.000	83.495
350.	97.000	251.000	97.476

The quarter-bridge results showed that measured strain and output voltage increased approximately linearly as the applied load increased. The strain calculated from the output voltage closely matched the displayed strain throughout the tested range. These results provided a baseline for comparing the output sensitivity of the half- and full-bridge configurations.

Table 2 Quarter Bridge

Load (g)	Strain Reading ($\mu\epsilon$)	Output (μV)	Calculated Strain ($\mu\epsilon$)
0.	0.000	0.000	0.000
50.	13.000	71.000	13.786
100.	27.000	142.000	27.573
150.	41.000	214.000	41.553
200.	55.000	284.000	55.146
250.	69.000	357.000	69.320
300.	83.000	429.000	83.301
350.	97.000	502.000	97.476

The half-bridge configuration produced approximately twice the output voltage of the quarter bridge while giving similar strain readings under the same loads. This showed that the additional active gauge increased the bridge's sensitivity, rather than increasing the deformation of the beam. The calculated strain also remained close to the displayed strain.

Table 3 Full Bridge

Load (g)	Strain Reading ($\mu\epsilon$)	Output (μV)	Calculated Strain ($\mu\epsilon$)
0.	0.000	0.000	0.000
50.	13.000	141.000	13.689
100.	27.000	283.000	27.476
150.	40.000	417.000	40.485
200.	54.000	557.000	54.078
250.	67.000	694.000	67.379
300.	81.000	841.000	81.650
350.	95.000	986.000	95.728

The full-bridge configuration produced the largest output voltage, approximately four times the quarter-bridge output, and twice the half-bridge output. However, its strain readings remained similar to those of the other configurations. This demonstrated that using all four active bridge arms increased the electrical response to strain without substantially changing the strain being measured. The calculated and displayed strain values also showed close agreement.

The graph shows that output voltage increased approximately linearly with load for all configurations, with the steeper full-bridge slope indicating greater sensitivity and the overlapping half-bridge curves showing the same response for both arrangements.

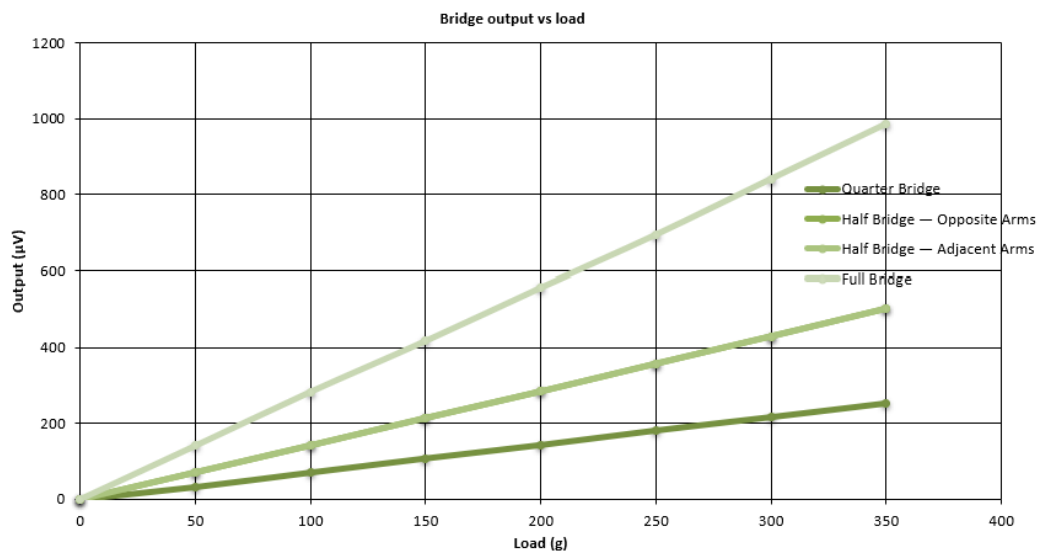


Figure 1 bridge output vs load

Table 4 Experiment 2 - The bending system

Load (g)	Force (N) = Load (Mass) $\times$ 9.81 m·s ⁻²	Strain Reading ($\mu\epsilon$)	Output Voltage (μV)	Bending Moment (Nm)	Calculated Stress (MN·m ⁻²) fix	Calculated Strain ($\mu\epsilon$)
0.	0.000	0.000	0.000	0.000	0.000	0.000
50.	0.491	12.000	32.000	0.199	2.384	11.516
100.	0.981	27.000	70.000	0.397	4.768	23.032
150.	1.472	41.000	107.000	0.596	7.151	34.548
200.	1.962	55.000	142.000	0.795	9.535	46.064
250.	2.453	69.000	180.000	0.993	11.919	57.580
300.	2.943	83.000	215.000	1.192	14.303	69.097
350.	3.434	97.000	251.000	1.391	16.687	80.613

Experiment 2 examined how increasing the load affected the beam's bending response. The bending moment, calculated stress, and measured strain increased approximately linearly with load. The measured Young's modulus was approximately 170 GPa, lower than the reference value of 207 GPa, consistent with the measured strain being generally higher than the theoretical strain.

The graph shows an approximately linear relationship between stress and strain, with its slope giving a measured Young's modulus of about 170 GPa, below the reference value of 207 GPa.

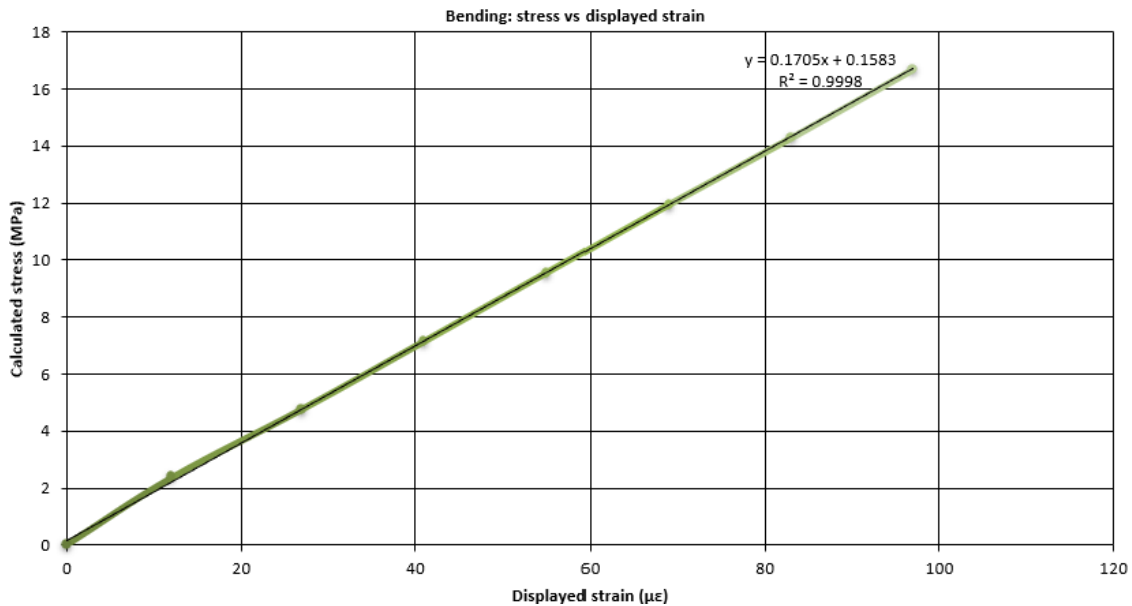


Figure 2 Bending: Stress vs. displayed strain

Table 5 Experiment 3: Individual Shear and Torque Gauges

Strain Gauge	Strain Reading ($\mu\epsilon$)	Polarity (+/-)	Type of Strain (Compressive/Tensile)
Blue	-25.000	-	Compressive
Red	23.000	+	Tensile
Yellow	-25.000	-	Compressive
Green	24.000	+	Tensile

The individual gauges recorded both tensile and compressive strains under torsion, with the red and green gauges showing positive readings and the blue and yellow gauges showing negative readings. Their similar magnitudes and opposite signs demonstrated the contrasting strain responses measured by the gauges during twisting.

Table 6 Experiment 3: Compare Strains

Load (kg)	Force (N) (Load $\times$ 9.81 m-s ⁻²)	Torque (Nm)	Output Voltage (μ V)	Strain Reading ($\mu\epsilon$)	Calculated (shear) Stress (MN-m ⁻²)	Calculated (direct) Strain ($\mu\epsilon$)
0.	0.000	0.000	0.000	0.000	0.000	0.000
0.25	2.453	0.368	127.000	12.000	1.874	11.769
0.5	4.905	0.736	254.000	24.000	3.747	23.537

As the applied torque increased, the measured strain increased proportionally and closely matched the calculated direct strain. This agreement showed that the shaft's measured response was consistent with the theoretical torsion calculations over the tested range.

The graph shows that measured and calculated direct strain increased proportionally with load, with closely aligned results indicating good agreement between the experimental readings and theoretical predictions.

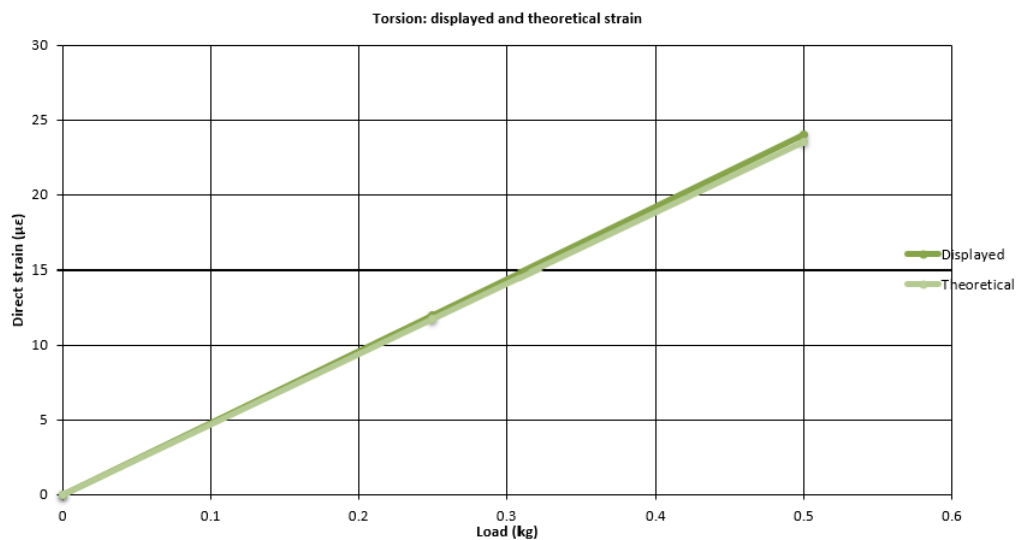


Figure 3 Torsion: displayed vs. theoretical strain

Table 7 Experiment 4: Red and Yellow Gauges

Load (kg)	Force (N)	Displayed Tensile Strain ($\mu\epsilon$)	Calculated Tensile Stress (KPa)	Calculated Tensile Strain ($\mu\epsilon$)
0.	0.000	0.000	0.000	0
1.	9.810	2.000	490.500	2.369565217
2.	19.620	6.000	981.000	4.739130435
3.	29.430	8.000	1471.500	7.108695652
4.	39.240	8.000	1962.000	9.47826087
5.	49.050	9.000	2452.500	11.84782609
6.	58.860	12.000	2943.000	14.2173913
7.	68.670	15.000	3433.500	16.58695652
8.	78.480	17.000	3924.000	18.95652174
9.	88.290	20.000	4414.500	21.32608696
10.	98.100		4905.000	23.69565217

The red and yellow gauges measured longitudinal tensile strain as the applied load increased. The measured strain generally increased with load, following the theoretical trend, although some readings showed uneven increases and differed from the calculated values.

The graph shows an overall increasing relationship between load and tensile strain, with the measured results generally falling below the theoretical line at higher loads.

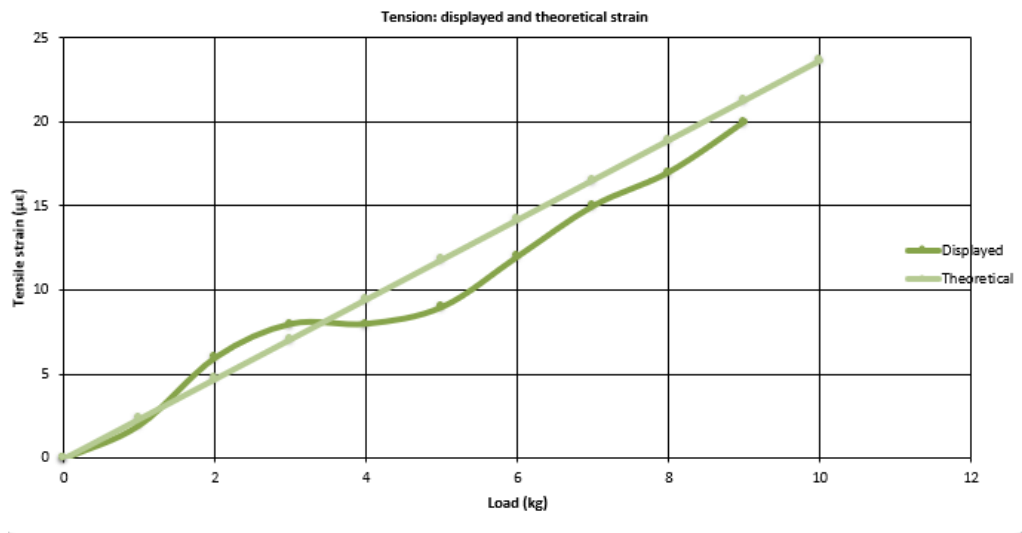


Figure 4 Torsion: Displayed vs theoretical strain

Table 8 Experiment 4: Blue and Green gauges

Load (kg)	Force (N)	Displayed Strain ($\mu\epsilon$)
0.	0.000	0.000
1.	9.810	-2.000
2.	19.620	-3.000
3.	29.430	-5.000
4.	39.240	-6.000
5.	49.050	-7.000
6.	58.860	-8.000
7.	68.670	-9.000
8.	78.480	-10.000

The blue and green gauges measured transverse strain as the tensile load increased. Their readings became more negative with increasing load, indicating greater sideways contraction as the specimen stretched lengthwise. This contraction is caused by the Poisson effect.

The colors identify the different gauge connections on the trainer. In the tension experiment, red and yellow measure longitudinal stretching, while blue and green measure transverse contraction. The negative values indicate shortening in the gauges' measurement direction, not that the applied load was negative.

The graph shows that transverse strain became increasingly negative as longitudinal strain increased, demonstrating the relationship between lengthwise stretching and sideways contraction.

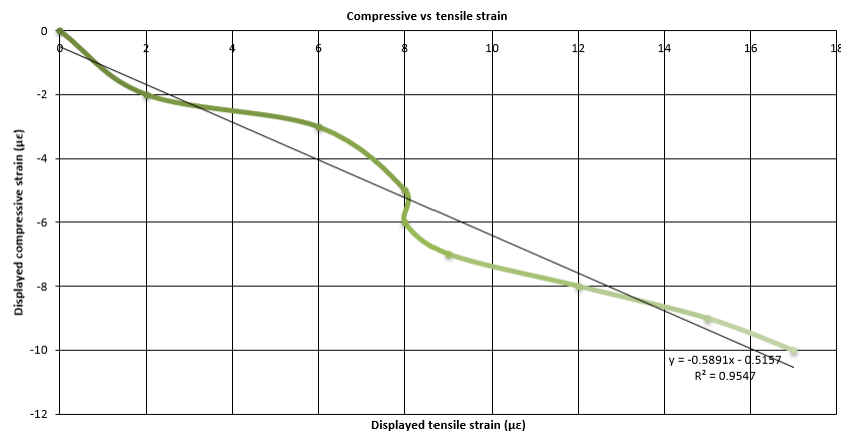


Figure 5 Compressive vs. tensile strain

Table 9 Experiment 4: Full Bridge

Load (kg)	Force (N)	Displayed Strain ($\mu\epsilon$)
0.	0.000	0.000
1.	9.810	2.000
2.	19.620	4.000
3.	29.430	7.000
4.	39.240	9.000
5.	49.050	12.000
6.	58.860	14.000
7.	68.670	17.000
8.	78.480	20.000

The full-bridge readings increased steadily as the tensile load increased, showing an approximately linear response. This configuration combined the longitudinal and transverse gauge responses into one bridge measurement.

The graph shows that displayed strain increased approximately linearly with load, indicating a consistent response over the tested range.

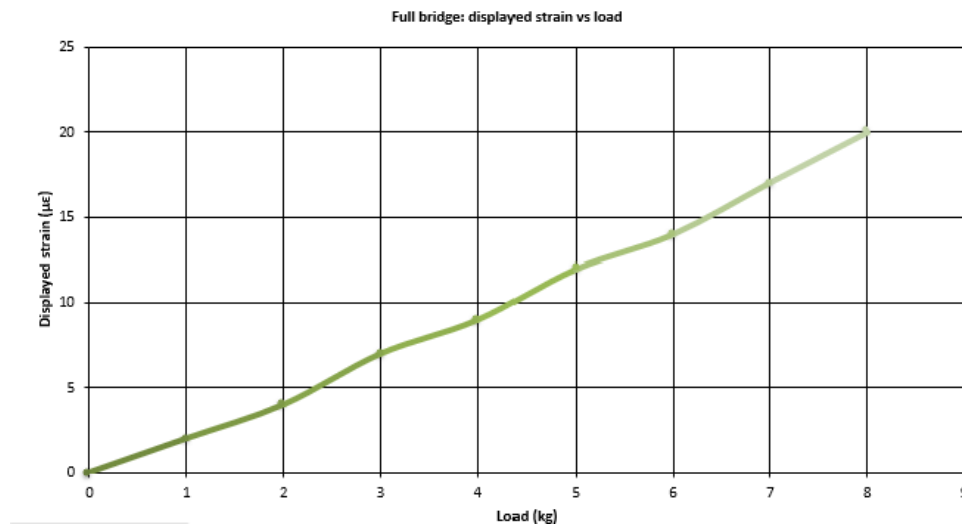


Figure 6 Experiment 4: Full Bridge

The equations used in the analysis connect the applied loads to the stresses and strains predicted for each specimen. The bridge equations relate output voltage to measured strain, while the bending, torsion, and tensile equations provide theoretical values for comparison. These relationships also support the determination of material properties, including Young's modulus and Poisson's ratio.

Conclusion

The experiment demonstrated how resistance strain gauges measure deformation under bending, torsion, and tensile loading. Comparing bridge configurations showed that increasing the number of

active gauges increased output sensitivity, while the measured strain remained similar under equivalent loads. The results generally showed an approximately linear relationship between applied load and strain.

The torsion measurements closely agreed with theoretical predictions, whereas the bending and tensile tests showed greater differences. The measured Young's modulus was approximately 170 GPa, below the reference value of 207 GPa. The tensile tests demonstrated longitudinal extension and transverse contraction, although the calculated Poisson's ratio requires verification of the bridge scaling before being accepted as a material property. Overall, the experiment connected electrical strain measurements with mechanics-of-materials calculations and highlighted the importance of correct bridge configuration and data interpretation.

References

- Al-Shudeifat, M. A., & Al-Khalil, K. S. (2019). Failure Analysis of Torsional Fatigue in Steel Shafts. *Engineering Failure Analysis*, 104, 1-12. Retrieved from <https://doi.org/10.1016/j.engfailanal.2019.104234>
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Appendices

For the additive, equal-magnitude strain arrangements used in Experiment 1:

$$\varepsilon \approx \frac{4V_o}{n GF V_s}$$

Here, V_o is the measured bridge output. Both voltages must be expressed in volts. This approximation applies to the Experiment 1 bridge arrangements, not automatically to every full-bridge arrangement.

Sample Calculations at 100 g

Quarter bridge: $V_o = 70 \mu V$, $n = 1$.

$$\varepsilon = \frac{4(70 \times 10^{-6})}{1(2.06)(5)} = 27.184 \times 10^{-6} = 27.184 \mu\varepsilon$$

Half bridge: $V_o = 142 \mu V$, $n = 2$. Both half-bridge tables have this reading.

$$\varepsilon = \frac{4(142 \times 10^{-6})}{2(2.06)(5)} = 27.573 \mu\varepsilon$$

Full bridge: $V_o = 283 \mu\text{V}$, $n = 4$.

$$\varepsilon = \frac{4(283 \times 10^{-6})}{4(2.06)(5)} = 27.476 \mu\varepsilon$$

These reproduce the calculated strain values in the Experiment 1 tables.

Experiment 2: Using example 100g row and the bending dimensions listed above.

1. Applied Force

$$F = mg$$

$$F = \left(\frac{100}{1000}\right)(9.81) = 0.981 \text{ N}$$

2. Second Moment of Area

For the rectangular beam:

$$I = \frac{bt_b^3}{12}$$

$$I = \frac{(0.020)(0.005)^3}{12} = 2.0833 \times 10^{-10} \text{ m}^4$$

3. Bending Moment at the Gauge

$$M = F(L - x)$$

$$M = (0.981)(0.420 - 0.015) = 0.397305 \text{ N m}$$

4. Surface Bending Stress

$$\sigma = \frac{My}{I}, y = \frac{t_b}{2}$$

$$\sigma = \frac{(0.397305)(0.0025)}{2.0833 \times 10^{-10}} = 4.76766 \times 10^6 \text{ Pa} = 4.768 \text{ MPa}$$

5. Theoretical Bending Strain

$$\varepsilon_{\text{theoretical}} = \frac{\sigma}{E_{\text{ref}}}$$

$$\varepsilon_{\text{theoretical}} = \frac{4.76766 \times 10^6}{207 \times 10^9} = 23.032 \times 10^{-6} = 23.032 \mu\varepsilon$$

The measured strain at this load was $27 \mu\varepsilon$.

6. Measured Young's Modulus

Young's modulus was determined from the best-fit slope of calculated stress against measured strain:

$$\sigma = E_{\text{measured}}\varepsilon + c$$

For an ordinary straight-line fit with an intercept:

$$E_{\text{measured}} = \frac{\sum_i (\varepsilon_i - \bar{\varepsilon})(\sigma_i - \bar{\sigma})}{\sum_i (\varepsilon_i - \bar{\varepsilon})^2}$$

Using the measured bending points, the spreadsheet reports:

$$E_{\text{measured}} = 170.523 \text{ GPa} \approx 170 \text{ GPa}$$

Strain must be dimensionless when calculating this slope. The reported modulus comes from the fitted data, not a single stress-to-strain ratio.

Experiment 3: Torsion calculations

Individual Gauge Readings

The individual-gauge test used $GF = 2.04$. The instrument directly displayed strains of -25 , $+23$, -25 , and $+24 \mu\varepsilon$ for blue, red, yellow, and green, respectively. These are measured inputs, not values calculated from shaft dimensions.

Sample Calculations at 0.50 kg

The following example uses the torsion comparison table, which lists $GF = 1.96$, $d = 0.010 \text{ m}$, $a = 0.150 \text{ m}$, and $G = 79.6 \text{ GPa}$.

1. Applied Force and Torque

$$F = mg$$

$$F = (0.50)(9.81) = 4.905 \text{ N}$$

$$T = Fa$$

$$T = (4.905)(0.150) = 0.73575 \text{ N m}$$

2. Polar Moment of Area

For a solid circular shaft:

$$J = \frac{\pi d^4}{32}$$

$$J = \frac{\pi(0.010)^4}{32} = 9.81748 \times 10^{-10} \text{ m}^4$$

3. Surface Shear Stress

$$\tau = \frac{Tr}{J}$$

$$\tau = \frac{(0.73575)(0.005)}{9.81748 \times 10^{-10}} = 3.74714 \times 10^6 \text{ Pa} = 3.747 \text{ MPa}$$

4. Shear Strain

$$\gamma = \frac{\tau}{G}$$

$$\gamma = \frac{3.74714 \times 10^6}{79.6 \times 10^9} = 47.075 \times 10^{-6}$$

5. Direct Strain at $\pm 45^\circ$

For gauges aligned at $\pm 45^\circ$ to the shaft axis under pure torsion:

$$\varepsilon_{\pm 45^\circ} = \pm \frac{\gamma}{2}$$

The magnitude is:

$$|\varepsilon| = \frac{47.075 \times 10^{-6}}{2} = 23.537 \mu\varepsilon$$

This reproduces the calculated direct strain in the table; the measured strain was $24 \mu\varepsilon$.

Experiment 4: Tensile Calculations

The following example uses the 8 kg row, for which both longitudinal and transverse readings are available.

1. Cross-Sectional Area

$$A = wt_s$$

$$A = (0.002)(0.010) = 2.00 \times 10^{-5} \text{ m}^2$$

2. Applied Force

$$F = mg$$

$$F = (8)(9.81) = 78.48 \text{ N}$$

3. Tensile Stress

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{78.48}{2.00 \times 10^{-5}} = 3.924 \times 10^6 \text{ Pa} = 3924 \text{ kPa}$$

4. Theoretical Longitudinal Strain

$$\varepsilon_L = \frac{\sigma}{E_{\text{ref}}}$$

$$\varepsilon_L = \frac{3.924 \times 10^6}{207 \times 10^9} = 18.957 \times 10^{-6} = 18.957 \mu\varepsilon$$

The red/yellow table recorded $17 \mu\varepsilon$, while the full-bridge table recorded $20 \mu\varepsilon$ at this load.

5. Poisson's Ratio—Single-Point Example

$$\nu = -\frac{\varepsilon_T}{\varepsilon_L}$$

Using the displayed longitudinal and transverse readings at 8 kg:

$$\nu = -\frac{-10 \mu\varepsilon}{17 \mu\varepsilon} = 0.588$$

The microstrain units cancel. This is a single-point illustration using the displayed readings.

6. Poisson's Ratio—Best-Fit Method

For transverse strain plotted against longitudinal strain:

$$\varepsilon_T = s\varepsilon_L + c, \nu_{\text{apparent}} = -s$$

$$s = \frac{\sum_i (\varepsilon_{L,i} - \bar{\varepsilon}_L)(\varepsilon_{T,i} - \bar{\varepsilon}_T)}{\sum_i (\varepsilon_{L,i} - \bar{\varepsilon}_L)^2}$$

The spreadsheet's fitted result is:

$$s = -0.58908, \nu_{\text{apparent}} = 0.58908 \approx 0.589$$

This reproduces the spreadsheet calculation. Treat it as an apparent experimental ratio until the instrument's scaling for the longitudinal and transverse bridge connections has been verified.

Full-Bridge Tensile Readings

Sheet 7 contains applied mass, calculated force, and instrument-displayed strain. Therefore, the force calculation above applies, but no separate voltage-to-strain sample calculation can be reproduced from that sheet because it does not list output voltages or the required conversion settings. Do not apply the Experiment 1 $n = 4$ equation automatically to this longitudinal/transverse arrangement.